

Colin Harfst

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Profile

Senior software engineer with 8+ years of experience developing cloud-native, full-stack web applications and delivering integrated software solutions. Proven track record of managing the end-to-end software lifecycle, leading technical initiatives within cross-functional product teams, and delivering operational improvements to drive business growth. Technical expertise across a variety of domains including: full-stack web development, APIs and microservices, scalable cloud architecture, DevOps, data engineering, and AI systems.

Work Experience

Senior Software Engineer | CarMax, Inc.

Aug 2023 – Present

- Designed and developed several scalable C#/.NET microservices to securely manage 500,000 customer documents per month in support of critical business operations
- Developed customer-facing and associate-facing full-stack web applications (C#/.NET, TypeScript, React) to upload, review, and transmit documents – eliminating manual intervention on 100,000 transactions per month
- Partnered with architects, engineers, and product managers to define roadmaps for re-platforming legacy applications – improving website reliability and security, while simplifying complex workflows
- Collaborated across product teams to define data contracts, establish SLAs, and triage incidents for customers and associates
- Designed an Azure Data Factory pipeline to synchronize data across systems; built a prototype using Azure AI Foundry and Document Intelligence to automate document review
- Developed and promoted reusable NuGet packages for shared data models and gateway connections
- Published and consumed enterprise events to support highly asynchronous business processes
- Owned end-to-end software lifecycle including CI/CD pipelines, Azure infrastructure provisioning, and production monitoring

Software Engineer II | CarMax, Inc.

Jun 2021 – Aug 2023

- Implemented, tested, and monitored numerous website enhancements to progress customers as they shop, research, and prepare to transact; developed and promoted reusable NPM packages for shared React components and hooks

Freelance Software Engineer | FlyCast River Forecast, LLC

Mar 2021 – Oct 2023

- Developed a full-stack Python/Django web application to register fly fishing hobbyists into a collective membership program, offering points-based rewards, daily weather reports with river conditions, and exclusive opportunities for guided trips

Senior Consultant | CapTech Ventures, Inc.

Jan 2021 – May 2021

- **Fortune 500 insurance holding company:** Developed a full-stack Python/Django web application for adjusters to track claims within an interactive dashboard; enhanced SQL ETL processes to normalize data from disparate source systems

Consultant | CapTech Ventures, Inc.

Aug 2018 – Jan 2021

- **Fortune 500 automotive retailer:** Developed a full-stack C#/.NET and React web application to replace a legacy sales system; integrated Azure SignalR to relay real-time financing offers from lenders to customers; designed a production monitoring dashboard using Azure App Insights; wrote automated end-to-end integration tests with WebdriverIO
- **Fortune 500 manufacturing company:** Developed a C#/.NET API for verifying user identities via integration with LexisNexis

Data Analyst Intern | IST Research Corp.

May 2017 – Aug 2017

- Developed a Python/Flask API to leverage Elasticsearch for reverse image searching and natural language processing (NLP) tasks (named entity recognition, sentiment analysis)

Education

University of Washington

Jan 2026 – Present

Professional Master's Certificate – Applied Cybersecurity Engineering

University of Virginia

Aug 2014 – May 2018

Bachelor of Arts – Mathematics and Computer Science

Skills & Technologies

Programming Languages: C#, Python, JavaScript, TypeScript, SQL, HTML, CSS, Java, C/C++

Frameworks and Tools: .NET, Entity Framework, Node.js, Express, React, Webpack, Vite, Django, Flask, Docker, Splunk

Cloud Services (Azure): App Services, Functions, Cosmos, SignalR, Service Bus, Redis, Key Vault, APIM, App Insights, Data Factory

Cloud Services (AWS): EC2, Lambda, S3, CloudWatch, RDS, DynamoDB

Artificial Intelligence: Agentic workflows with OpenCode and GitHub Copilot, Custom machine learning models

Additional Work and Research Experience: Undergraduate Teaching Assistant, UVA CS Dept. – Discrete Mathematics, 3 semesters | Undergraduate Researcher, UVA Mathematics Dept. – Theoretical and applied hypergraph research using Macaulay2